REVISION 2021 • SEMESTER 2 • ENGINEERING SCIENCE

Manufacturing Technology

A complete diploma engineering handbook for Course Code **2022**. This lesson converts the official syllabus into a practical manufacturing textbook covering foundry, casting, metal working, powder metallurgy, welding, brazing, soldering, forging and press working.

COURSE CODE
2022

COURSE TITLE
Manufacturing Technology

PROGRAMME
Mechanical Engineering / Mechatronics

SEMESTER
2

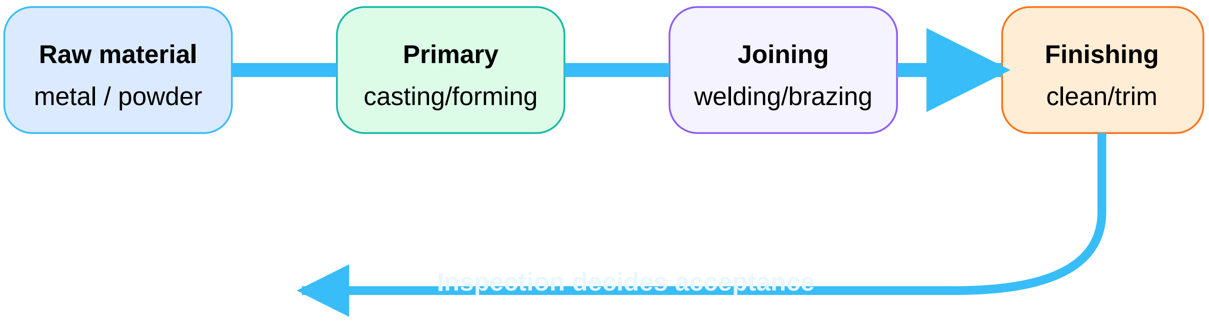
CREDITS
3

CATEGORY
Engineering Science

PERIODS/WEEK
3 (L:3 T:0 P:0)

PERIODS/SEMESTER
45

From raw material to finished product



Manufacturing technology connects drawing, material, process selection, tooling, inspection, safety and cost.

COURSE OVERVIEW

What this subject teaches

Manufacturing converts raw stock into useful products by changing shape, joining material or improving final form. For a diploma engineer, this is not just theory. It decides whether a part can be made repeatedly, safely, economically and with correct quality.

Why it matters

Every machine frame, bracket, pulley, casting, welded support, punched sheet and forged component needs process knowledge before design can become a real product.

Objectives

- Understand importance of manufacturing processes.
- Study basic casting, welding and forming methods.
- Select suitable process for a given part.

Prerequisite

Engineering Workshop Practice: basic manufacturing process, foundry, welding etc. from Semester 1 and 2.

Malayalam support

> Manufacturing Technology □□□□□□□ raw material □□□ usable product □□□□□□□□□□□ practical process □□□□□□□□. Exam-□ process principle, labelled diagram, operation sequence, advantages, limitations, defects and applications □□□□□□□.

COURSE OUTCOMES

Official outcome structure

CO	Outcome	Hours	Cognitive level	Engineering meaning
CO1	Explain manufacturing process and relevance of foundry.	11	Understanding	Select pattern, moulding sand, gating and riser basics.
CO2	Identify and explain casting and metal working processes.	10	Understanding	Compare casting types, cold/hot working and powder metallurgy.
CO3	Describe metal joining process and applications.	12	Understanding	Choose welding/brazing/soldering method and identify defects.
CO4	Explain forging and press working concepts.	10	Understanding	Recognise forging tools, press operations and die-set parts.

TABLE OF CONTENTS

Clickable handbook sections

SOURCE AND SYLLABUS INFORMATION

Confirmed official information

Item	Status from supplied syllabus/source
Programme	Diploma in Mechanical Engineering / Mechatronics
Course code and title	2022 - Manufacturing Technology
Semester / credits / category	Semester 2 / 3 credits / Engineering Science
Periods	3 periods per week, 45 periods per semester
CO-PO mapping	CO1-PO1=2; CO2-PO1=2; CO3-PO1=2 and PO4=2; CO4-PO1=2
Official model question paper pattern	Not specified in supplied syllabus/source.

REDESIGNED SYLLABUS ROADMAP

What to learn, draw, explain and apply

M1 Foundry Basics

Hours: 11. **CO:** CO1.

Draw: pattern, mould, gating/riser. **Explain:** properties, sand, allowances. **Exam importance:** very high.

Common mistake: mixing pattern with mould.

M2 Casting + Working

Hours: 10. **CO:** CO2.

Draw: centrifugal/die casting and powder metallurgy flow. **Explain:** defects, hot/cold working.

Common mistake: writing cold and hot working without temperature basis.

M3 Joining

Hours: 12. **CO:** CO3.

Draw: welding classification, flames, arc welding, spot welding, joint types. **Explain:** electrodes, positions, defects.

Common mistake: confusing brazing and soldering limits.

M4 Forging + Press

Hours: 10. **CO:** CO4.

Draw: forging tools, press tool, punching vs blanking. **Explain:** tools, operations, clearance effect.

Common mistake: interchanging punching and blanking product.

MODULE 1 • CO1 • 11 HOURS

Manufacturing Process and Foundry Fundamentals

Learn material properties, pattern types, pattern materials, allowances, moulding sand, moulding methods, gating system, risers, chillers and casting cleaning.

Manufacturing process

Manufacturing changes raw material into useful product through casting, forming, machining, joining and finishing. Selection depends on material, shape, quantity, accuracy, surface finish and cost.

> Process selection wrong □□□□□□□□ design correct □□□□□□ part manufacture □□□□□□□ □□□□□□□□□□□□□□□.

Mechanical properties

Strength resists load; hardness resists indentation; brittleness means sudden fracture; toughness absorbs impact energy; malleability allows sheet formation; ductility allows wire drawing; stiffness resists elastic deformation; fatigue is failure by repeated load; resilience stores elastic energy.

Patterns and materials

Single piece, split, match plate, gated, loose piece and sweep patterns are used according to casting shape and production rate. Wood is cheap but less durable; metal is accurate and durable; plastic is light; plaster gives fine detail but is brittle.

Pattern allowances

Shrinkage allowance compensates solidification contraction, draft allowance permits easy withdrawal, machining allowance gives extra stock, distortion/camber allowance offsets warping and rapping allowance compensates cavity enlargement.

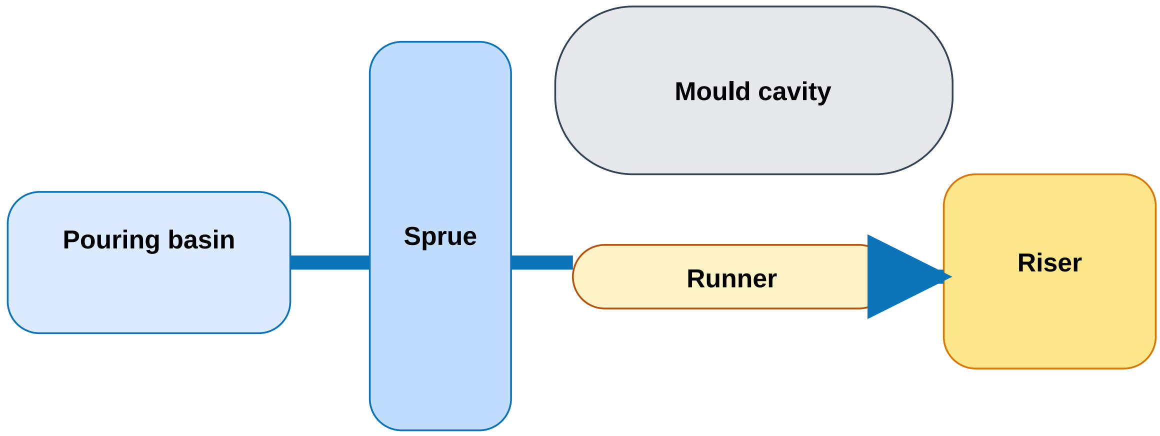
Moulding sand

Good sand needs porosity, plasticity, adhesiveness, cohesiveness and refractoriness. Types: green sand, dry sand, parting sand, loam sand, facing sand and core sand. Chaplets support cores inside the mould cavity.

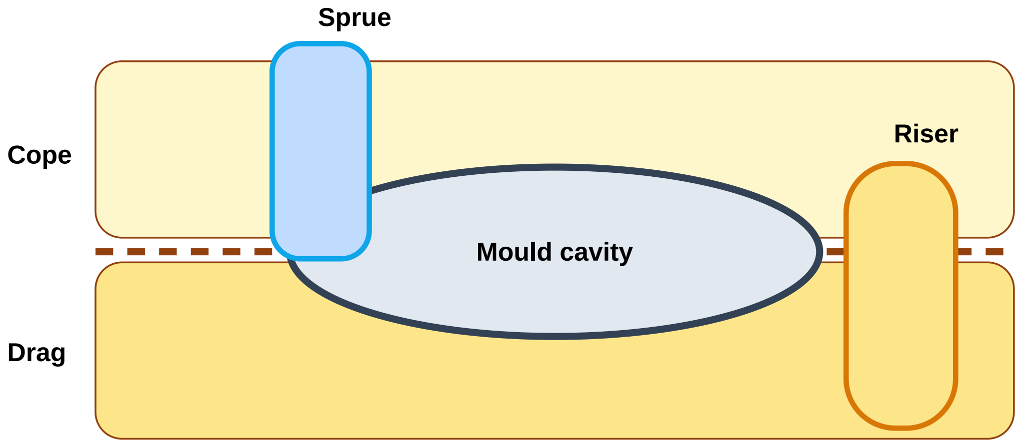
Moulding methods

Bench moulding is for small work, floor moulding for medium castings, pit moulding for large/heavy castings and sweep moulding for symmetrical large cavities.

Engineering diagram lab



Gating system and riser: draw all labels clearly.



Sand mould with cope, drag, parting line, sprue and riser.

Practical / industrial relevance

Foundry is used for pump housings, machine frames, brackets, pulleys, heavy bases and complex parts difficult to machine from solid stock. In escalator/elevator work, cast and fabricated components must be checked for cracks, blow holes, poor finish and wrong dimensions.

Expected questions

- Define manufacturing process and list classifications.

- Explain any five mechanical properties.
- Compare pattern materials.
- Explain pattern allowances.
- Draw gating system and explain riser functions.

MODULE 2 • CO2 • 10 HOURS

Casting Processes, Metal Working and Powder Metallurgy

Study casting types, defects, cold working, hot working and elementary powder metallurgy.

Casting categories

Castings may be ferrous, non-ferrous or special. Ferrous casting uses iron/steel alloys; non-ferrous casting uses aluminium, copper, zinc and similar alloys; special casting uses controlled processes for better surface finish, accuracy or material saving.

Centrifugal and die casting

Centrifugal casting rotates the mould and uses centrifugal force to feed metal. Die casting forces molten metal into metal dies. Gravity die casting uses gravity filling; pressure die casting uses external pressure. Hot chamber machines use goose-neck/direct injection; cold chamber machines are used for higher melting alloys.

Slush and vacuum casting

Slush casting forms hollow thin products by allowing partial solidification near the mould wall and pouring out excess metal. Vacuum casting removes air/gases and helps reduce porosity.

Casting defects

Common defects: blow holes, shrinkage cavity, cold shut, misrun, hot tears, sand inclusion and mismatch. Causes include gas entrapment, low temperature, poor gating, wrong sand and bad mould alignment.

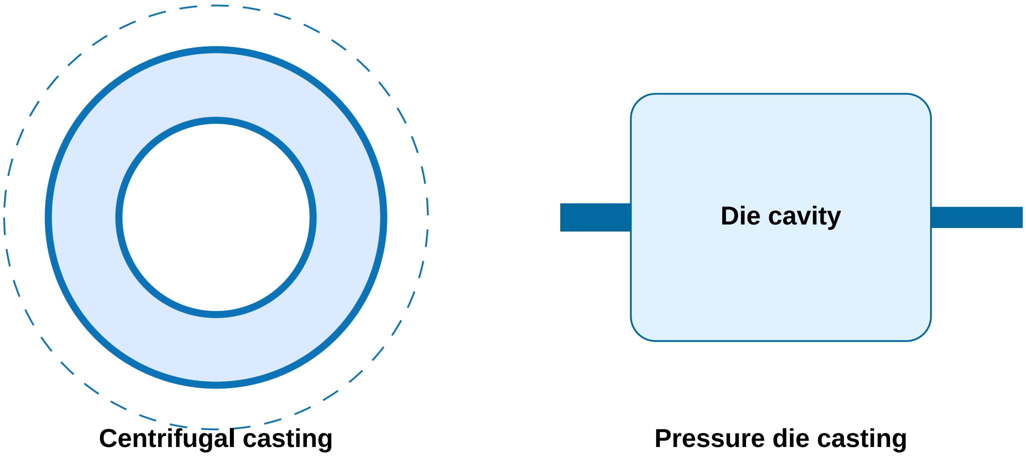
Cold and hot working

Cold working is done below recrystallisation temperature: good finish, high accuracy, strain hardening but high force. Hot working is done above recrystallisation temperature: lower force, grain refinement and less work hardening, but scaling and poorer finish.

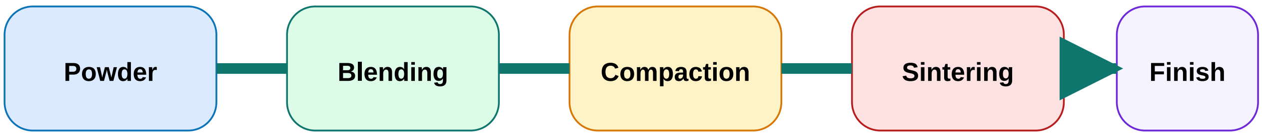
Powder metallurgy

Stages: powder production, blending, compaction, sintering and finishing. Advantages: near-net shape, less waste, porous parts possible and ability to process high melting point materials.

Engineering diagram lab



Centrifugal casting and pressure die casting concept.



Powder metallurgy process flow.

Troubleshooting table

Defect	Probable cause	Control
Blow holes	Gas / moisture	Proper venting and sand control
Misrun	Low fluidity or low temperature	Correct pouring temperature and gating
Cold shut	Two metal streams do not fuse	Improve design, temperature and feed
Shrinkage	Insufficient feeding	Correct riser and directional solidification

Expected questions

- Explain centrifugal casting with neat sketch.
- Compare hot working and cold working.
- List casting defects and remedies.
- Explain powder metallurgy stages and applications.
- Differentiate gravity die and pressure die casting.

MODULE 3 • CO3 • 12 HOURS

Metal Joining: Welding, Brazing and Soldering

Understand joining classification, gas welding, flames, arc welding, resistance welding, thermit welding, electrodes, welding positions, joints, brazing and soldering.

Classification of welding

Welding is broadly classified as fusion welding and pressure welding. Fusion welding melts the base metal or filler metal. Pressure welding applies heat and pressure or pressure alone to produce coalescence.

Gas welding and flames

Gas welding uses oxy-acetylene flame. Neutral flame is general purpose, carburizing flame has excess acetylene and oxidizing flame has excess oxygen. Flame selection affects weld metal chemistry.

Arc welding

Arc welding uses electric arc between electrode and workpiece. Equipment includes AC transformer or DC generator, electrode holder, cables, earth clamp and protective gear. SMAW, SAW, TIG and MIG are important arc processes.

Resistance and thermit welding

Resistance spot welding joins sheets by pressure and heat from electrical resistance. Seam welding makes continuous leak-proof seam. Projection welding localises current at projections. Thermit welding uses exothermic reaction heat.

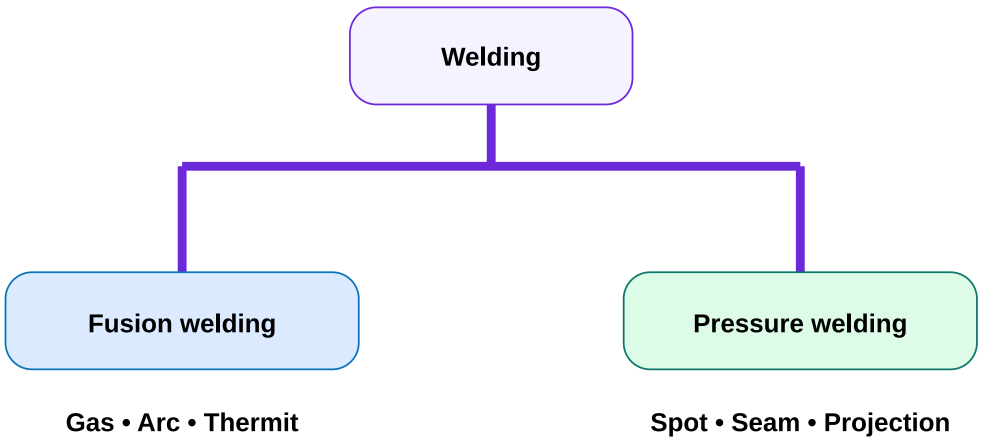
Electrodes and positions

Electrode coating stabilizes arc, shields molten metal, forms slag, deoxidizes and may add alloying elements. Positions: flat, horizontal, vertical and overhead.

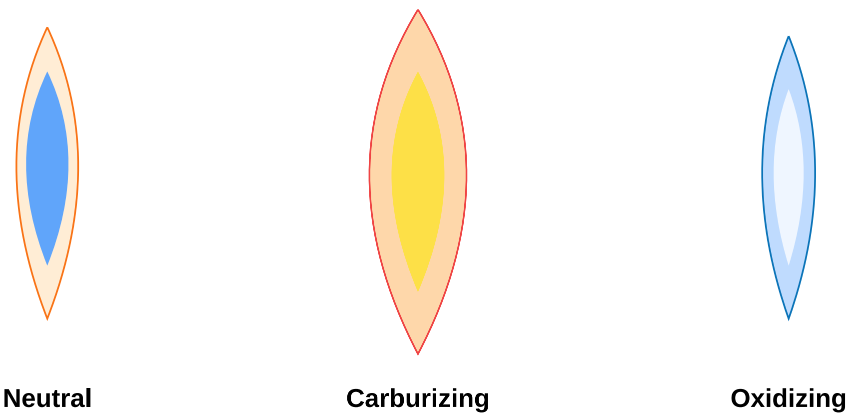
Brazing and soldering

Brazing uses filler metal above 450 deg C and below base metal melting point. Soldering uses filler below 450 deg C. Both need clean surface, flux and proper joint clearance.

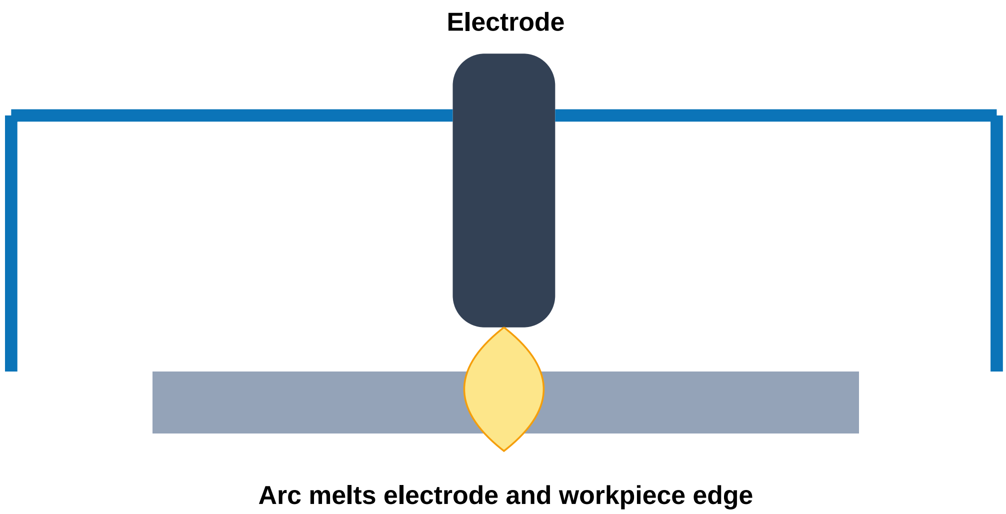
Engineering diagram lab



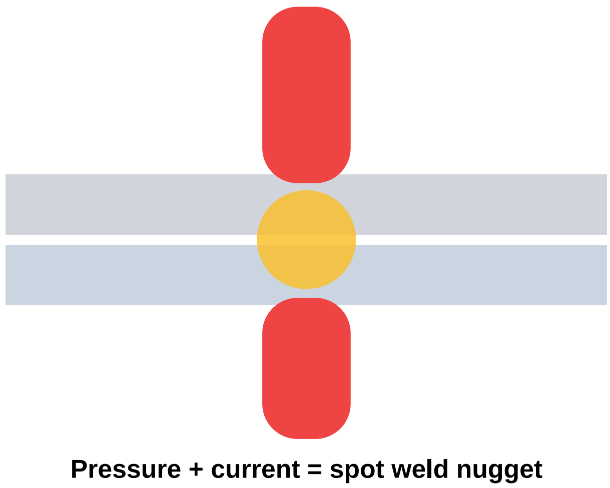
Welding classification.



Three oxy-acetylene welding flames.



Arc welding principle.



Resistance spot welding.

Welding defects

Porosity, slag inclusion, undercut, incomplete penetration, cracks, distortion and spatter. Remedies: clean surface, correct current, correct travel speed, proper electrode angle, correct edge preparation and preheat where needed.

Expected questions

- Classify welding processes.
- Explain gas welding flames.
- Draw arc welding setup.
- Explain functions of electrode coating.
- Compare brazing and soldering.

MODULE 4 • CO4 • 10 HOURS

Forging and Press Working

Learn forging process classification, forging tools, press working machines, press operations, die-set parts and punching/blanking comparison. No numerical problems are prescribed.

Forging processes

Forging shapes metal by compressive force, normally using hammer or press. Flat die/open die forging is done between simple dies. Closed die forging uses shaped die cavity to produce accurate parts.

Forging tools

Anvil supports work, swage block supports shaped work, hammers apply impact, tongs hold hot work, chisels cut, swages/fullers form sections, flatter smooths, set hammer makes shoulders, punch makes holes and drift enlarges holes.

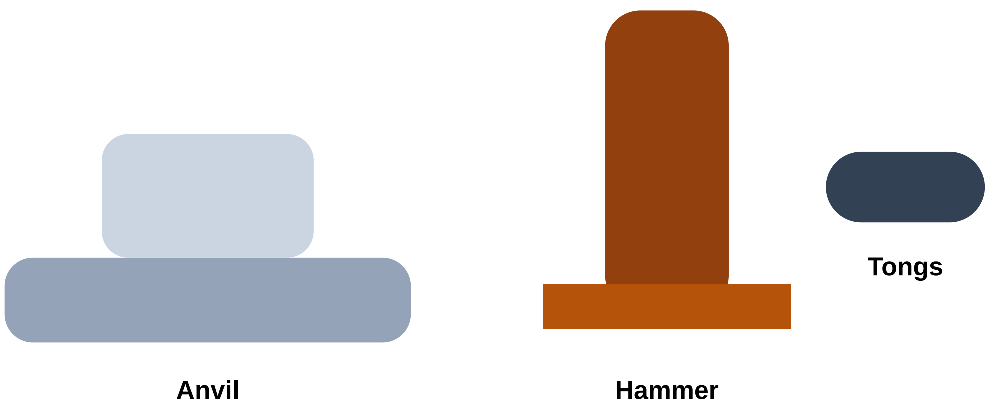
Press working machines

Presses use ram and die to work sheet metal. Important specifications: capacity/tonnage, stroke, shut height, bed size, speed and type of drive.

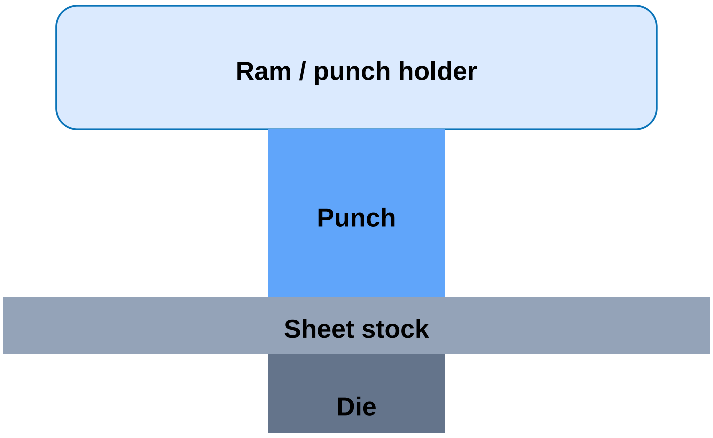
Press operations

Cutting, bending, drawing, punching, blanking, notching and lancing are common operations. Clearance between punch and die affects force, burr, edge quality and tool life.

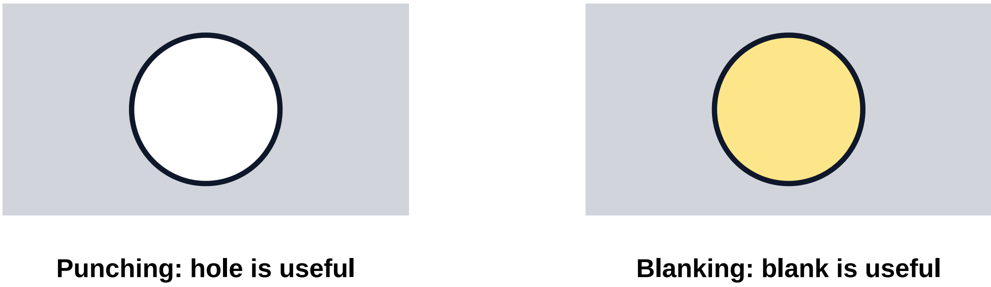
Engineering diagram lab



Basic forging tools.



Press tool principle.



Punching versus blanking.

Industrial relevance

Forging is used for high strength parts like shafts, levers, connecting rods and tools. Press working is used for sheet metal covers, brackets, washers, slots, louvers, notches, control panel holes and production components.

Expected questions

- Classify forging processes.
- Explain forging tools and uses.
- Explain press operations.
- Compare punching and blanking.
- Explain die-set components and clearance effect.

FORMULA / RULE BANK

Rules, definitions and diagram checklist

Shrinkage allowance

Rule: Pattern is made larger than final casting to compensate contraction during solidification and cooling.

Used in: Pattern design.

Draft allowance

Rule: Taper is provided on vertical faces so pattern can be withdrawn without damaging mould.

Used in: Sand moulding.

Machining allowance

Rule: Extra metal is left on casting surface for finishing by machining.

Used in: Castings requiring accurate surfaces.

Recrystallisation temperature

Rule: Hot working is above it; cold working is below it. This is the correct basis for comparison.

Brazing vs soldering

Rule: Brazing filler melts above 450 deg C; soldering filler melts below 450 deg C. Base metal is not melted in either case.

Punching vs blanking

Rule: In punching, hole is useful and slug is scrap. In blanking, blank is useful and remaining strip is scrap.

Diagram checklist

Every manufacturing answer should include title, process principle, neat labelled sketch, sequence, advantages, limitations, applications and defects/safety where relevant.

Safety bank

Use PPE, goggles, gloves, apron, welding shield, tongs for hot work, fume extraction, proper earthing and machine guards. Never touch hot metal directly.

SOLVED EXAMPLES / PROBLEM LAB

Answer-writing examples with full steps

Example 1: Choose process for a pump housing

Given: Complex internal cavity, medium production, cast iron.

Logic: Machining from solid wastes material; forging cannot make internal cavity easily. Sand casting allows complex shapes using cores.

Answer: Use sand casting with core, proper gating and riser. Inspect for blow holes and shrinkage.

Exam tip: write process selection reason, not only process name.

Example 2: Identify hot/cold working

Question: Rolling of heated steel billet above recrystallisation temperature.

Reason: Temperature is above recrystallisation; metal deforms with lower force and grain refinement.

Answer: Hot working. Advantage: lower force and improved grain structure. Limitation: scale and poor finish.

Example 3: Welding defect remedy

Problem: Porosity appears in weld bead.

Cause: Moisture, dirty surface, long arc, wrong shielding.

Remedy: Clean joint, dry electrode, correct arc length and shielding gas/flux.

Example 4: Press operation choice

Given: A round washer must be produced from sheet.

Logic: Outer disc removal is blanking; inner hole removal is punching.

Answer: Use blanking for outside profile and punching for centre hole.

QUICK REVISION

Last-day revision and practice

One-line revision points

- Pattern is the replica used to make mould cavity.
- Riser feeds molten metal during shrinkage.
- Chiller increases local cooling rate.
- Cold working improves finish and strength but increases force.
- Hot working needs lower force but gives scale.
- Electrode coating shields, stabilizes and forms slag.
- Brazing and soldering do not melt base metal.
- Punching creates useful hole; blanking creates useful blank.

Common exam traps

- Do not write pattern allowances as sand properties.

- Do not forget diagram labels in gating/welding/press questions.
- Do not say all welding melts base metal; pressure welding may not.
- Do not confuse die casting with sand casting.
- Do not reverse punching and blanking.

Objective practice

1. Riser is used to

A. feed shrinkage
B. hold electrode
C. increase punch clearance
D. cut sheet

Answer: A

2. Cold working is done

A. above melting point
B. below recrystallisation temperature
C. only in furnace
D. only by casting

Answer: B

3. Spot welding is a type of

A. gas welding
B. resistance welding
C. thermit welding
D. soldering

Answer: B

4. In blanking, the useful part is

A. hole
B. slug/blank
C. riser
D. runner

Answer: B

Module-wise expected questions

M1: Explain pattern allowances with examples.

M1: Draw and explain gating system and riser.

M2: Explain centrifugal casting and die casting.

M2: Compare hot working and cold working.

M3: Classify welding and explain arc welding.

M3: Explain electrode coating functions and welding defects.

M4: Explain forging tools.

M4: Compare punching and blanking with sketch.

FRESH MODEL QUESTION PAPER

Manufacturing Technology 2022 - Practice Model Paper

Note: The supplied syllabus does not provide the official model question paper pattern. This fresh practice paper is arranged for full module and CO coverage; it does not copy official questions.

Course Code: 2022
Course: Manufacturing Technology
Time: 3 Hours
Maximum Marks: 100

Part A - Short answer questions
1. Define manufacturing process and list two objectives.
2. Write any four mechanical properties of engineering materials.
3. State the functions of runner, gate and riser.
4. List four types of patterns.
5. Define cold working and hot working.
6. Name four casting defects.
7. What is powder metallurgy?
8. List welding positions.
9. Differentiate brazing and soldering.
10. Define punching and blanking.

Part B - Descriptive questions
11. Explain pattern materials and pattern allowances.
12. Draw a sand mould and explain moulding sand properties.
13. Explain centrifugal casting and pressure die casting.
14. Compare hot working and cold working with examples.
15. Explain powder metallurgy stages, advantages and applications.
16. Classify welding processes with neat chart.

17. Explain gas welding flames and applications.
18. Draw arc welding setup and explain electrode coating functions.
19. Explain resistance welding methods and welding defects.
20. Explain forging tools and press working operations.
21. Draw and explain die-set components. Compare punching and blanking.

Part C - Application / diagram questions
22. Select suitable manufacturing process for a complex pump housing and justify.
23. A sheet-metal control panel requires circular holes and rectangular slots. Identify press operations required and explain.
24. A weld bead has porosity and slag inclusion. State causes and remedies.

COMPLETE MODEL ANSWERS

Full answer guide

Answers 1-4

Manufacturing is conversion of raw material into useful product by changing shape, size, surface, property or joining condition. Properties: strength, hardness, brittleness, toughness, malleability, ductility, stiffness, fatigue and resilience. Runner distributes molten metal, gate controls entry and riser feeds shrinkage. Pattern types include single piece, split, match plate, gated, loose piece and sweep.

Answers 5-10

Cold working is below recrystallisation temperature; hot working is above it. Defects: blow holes, shrinkage, cold shut, misrun, hot tear, sand inclusion. Powder metallurgy uses metal powders compacted and sintered. Positions: flat, horizontal, vertical, overhead. Brazing uses filler above 450 deg C; soldering below 450 deg C. Punching creates hole; blanking creates useful blank.

Answers 11-12

Pattern materials: wood, metal, plastic and plaster with advantages and limitations. Allowances: shrinkage, draft, machining, distortion/camber and rapping. Moulding sand properties: porosity, plasticity, adhesiveness, cohesiveness and refractoriness. Draw cope, drag, cavity, parting line, sprue and riser.

Answers 13-15

Centrifugal casting uses rotation; pressure die casting injects molten metal into die. Hot working needs lower force and refines grains; cold working gives better finish and strain hardening. Powder metallurgy stages: powder production, blending, compaction, sintering and finishing; applications include bearings, filters, gears and electrical contacts.

Answers 16-19

Welding is fusion or pressure welding. Gas welding flames: neutral, carburizing and oxidizing. Arc welding uses electrode, power source, cables, holder, work and arc. Electrode coating stabilizes arc, shields weld, forms slag and deoxidizes. Resistance welding includes spot, seam and projection. Defects: porosity, slag inclusion, cracks, undercut and incomplete penetration.

Answers 20-24

Forging tools: anvil, swage block, hammers, tongs, chisels, swages, fullers, flatter, set hammer, punch and drift. Press operations: cutting, bending, drawing, punching, blanking, notching and lancing. Die-set parts: punch, die, die shoe, guide pin, bolster plate, stripper, stock guide, feed stock and pilot. Pump housing suits sand casting with core. Control panel holes need punching and slots/notches. Weld porosity/slag need clean surface, correct current, dry electrode and suitable speed.

REFERENCES

Official references and supplementary links

Type	Reference
T1	Manufacturing process - Begeman - 5th Edition - McGraw Hill, New Delhi 1981
R1	Workshop Technology - W A J Chapman - Volume I, II & III - Vima Books Pvt. Ltd.
R2	Production Technology - HMT - Edn. 18 - Tata McGraw Hill Publishing Co. Ltd. - 2001
R3	Production Technology - P. C. Sharma - Edn. X - S. Chand & Co. Ltd. - 2006
R4	Production Technology, Edn. XII, Khanna Publishers - 2006
Online 1	https://nptel.ac.in/courses/112107144/
Online 2	https://nptel.ac.in/courses/112104195/